

Paper Selection Proposal

Single-Source Shortest Paths with Negative Real Weights in $\tilde{O}(mn^{8/9})$ Time

Team Members: Hussain Umer Farooqui, Hussain Mustansir

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Paper Details

- **Title:** Single-Source Shortest Paths with Negative Real Weights in $\tilde{O}(mn^{8/9})$ Time
- **Authors:** Jeremy T. Fineman
- **Conference:** Submitted to STOC (2024)
- **Year:** 2024
- **DOI/Link:** <https://dl.acm.org/doi/10.1145/3618260.3649614>

Summary

This paper presents a randomized algorithm for solving the Single-Source Shortest Paths (SSSP) problem on directed graphs with real edge weights, both positive and negative. The key contribution is an improved running time of $\tilde{O}(mn^{8/9})$, where m is the number of edges and n the number of vertices, which improves upon the Bellman-Ford algorithm's $O(mn)$ time by using Dijkstra Algorithm.

Justification

This paper's novel time complexity improvement for SSSP is relevant to graph-based optimization problems, including network analysis and routing. It offers a practical improvement for graphs with real weights and negative edges. And it aligns with one of the topics in our course syllabus.

Implementation Feasibility

The paper provides detailed theoretical descriptions of the algorithm. While no direct code is available, existing algorithms like Bellman-Ford and Dijkstra can serve as a foundation for implementation.

Team Responsibilities

Both of us will contribute equally to every task.